30 MAR 2026

25FEBS201IAE2

Seat No. : 35

Student Name : Prachi

St. John College of Engineering and Management, Palghar,
Maharashtra
Autonomous Institute

Internal Assessment Examinations, March 2026

ASH - SEMESTER II

25FEBS201 - Engineering Mathematics II

Time : 1 Hour **Maximum Marks : 20**

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Max. marks(5)	Course Outcome	Bloom's Level
	a)What is the value of the double integral $\int_0^1 \int_0^x x\, dydx$?	1	CO4	L1
	b)Find the value of the integral $\int_0^\pi \int_0^\pi \int_0^\pi sin(x)\, dxdydz$	1	CO5	L1
	c)In Euler's method, what does h represent?	1	CO6	L1
	d)What is the integral of $\int \frac{1}{x^2+a^2}\, dx$?	1	CO4	L1
	e)Write the conversion formula from cartesian to spherical coordinates for x, y, z.	1	CO5	L1
	f)What is the formula for Simpson's 3/8th Rule?	1	CO6	L1
Q2	a)Change the order of integration $\int_0^\infty \int_0^x f(x,y)dydx$	5	CO4	L3
	OR			
	b)Evaluate $\int_0^1 \int_0^x (x^2+y^2)xdydx$	5	CO4	L3
Q3	a)Evaluate $\int_0^2 \int_1^2 \int_0^{yz} xyzdxdydz$	5	CO5	L3
	OR			
	b)Find the area bounded by parabola $y^2 = 4x$ and the line $y = 2x - 4$.	5	CO5	L3

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Q4	a)Solve the following using Eulers Method, $\frac{dy}{dx} = x^2 + y^2, y(0) = 1$ find y at $x = 1$ in 5 steps.	5	CO6	L3
	OR			
	b)Solve the following using Modified Eulers Method, $\frac{dy}{dx} = x + 3y, y(0) = 1$ find y for $x = 0.05$. Correct to four decimal places	5	CO6	L3

